

When Do Agent Harness Improvements Generalize?

Semantic Binding, External Validity, and Evaluation Reliability in LLM Agents

Under review as a conference paper at ICLR 2027

Abstract

A harness intervention—the way a task presents its evidence and disambiguating context to an agent—can survive repeated development experiments and a pre-frozen within-family held-out test, yet disappear under a new model or a new semantic task family. We study this on tool-grounded EDA workflows. On a controlled *semantic-handoff* task, a non-answer-bearing “clarity bundle” (BundleS: canonical role labels, value-domain definitions, a glossary, a procedural contract; no golden values) suppresses a specific semantic-axis binding failure for one frontier model, and this replicates on a pre-frozen held-out instance. Under exact counterbalancing the benefit is *not established* for a second model; on two structurally independent task families it is a clean null in one family and non-discriminative at ceiling in the other; a machine-readable typed representation of the same information adds no measurable advantage. The central finding is that **within-family held-out confirmation did not imply cross-model or cross-family transfer**. With three primary instances per external family, our design is diagnostic rather than population-estimating: it tests existence, recurrence, ceiling/null behavior, and transfer direction; it cannot resolve small effect sizes or support population-level success-rate estimates. Because each conclusion could be manufactured or masked by an evaluation artifact, we additionally contribute a reusable *Harness Effect Audit Protocol*—four layers (capability validity, sampling validity, execution validity, artifact integrity) operationalized as nine controls—each shown catching a real threat. The contribution is an evaluation study of harness effects, not a proposal of an improved harness.

1 Introduction

A harness intervention can survive repeated development experiments and a pre-frozen within-family held-out test, yet disappear under a new model or a new semantic task family. This is the result that motivates the paper. It is uncomfortable for agent evaluation: an effect that looks robust under repeated within-condition replication and even under a frozen held-out instance can still fail the first strict external-validity test. If the community takes a single within-family replication as evidence that a harness improvement “works,” it will systematically over-credit interventions that are model- or family-specific.

We study this on tool-grounded EDA workflows, where an LLM agent reads heterogeneous task evidence, binds quantities to canonical typed roles, and submits an artifact to a real commercial tool for signoff. We focus on a *semantic-handoff* task: the agent must assign a tuple of values to canonical typed roles (e.g. place a PVT descriptor on the correct typed axis) where the visible evidence uses misleading role vocabulary. A wrong binding is not a syntax error—it produces a *plausible green signoff* or *plausible numeric output* on the real tool. Only a typed provenance/authority grader distinguishes a correct from a wrong binding, making the task a clean instrument for whether harness information structure changes *semantic execution behavior* as opposed to mere tool success. We ask:

- **RQ1.** Can harness information structure change semantic execution behavior?
- **RQ2.** Does an observed harness effect transfer across models and task families?
- **RQ3.** Which evaluation controls are required for those conclusions to be valid?

In brief: (RQ1) yes—within a family a non-answer-bearing clarity bundle suppresses a specific axis-binding failure for one model, confirmed on a pre-frozen held-out instance; (RQ2) this does not imply transfer—it is not established for a second model under counterbalancing, and is a clean null

or a ceiling on two independent families; (RQ3) the conclusions depend on transport, tool health, sample-membership arbitration, protocol completion, action-surface, and canonical-source integrity, each able to manufacture or mask a result.

Contributions. (1) A controlled *semantic-binding* mechanism with a typed failure taxonomy, isolated from analog-design difficulty: the failure (placing a value on the wrong typed axis) stays tool-green and is rejected only by a typed provenance/authority grader. (2) The *external-validity-failure-after-held-out-confirmation* finding: a within-family effect confirmed on a pre-frozen held-out instance did not transfer across model or across two structurally independent families; a typed representation added no advantage. (3) A reusable **Harness Effect Audit Protocol**—four layers, nine controls—each shown catching a real threat to the headline interpretation. The paper is organized as three studies mapped to the three questions; the repository chronology is an appendix provenance record. Figure 1 previews the study pipeline.

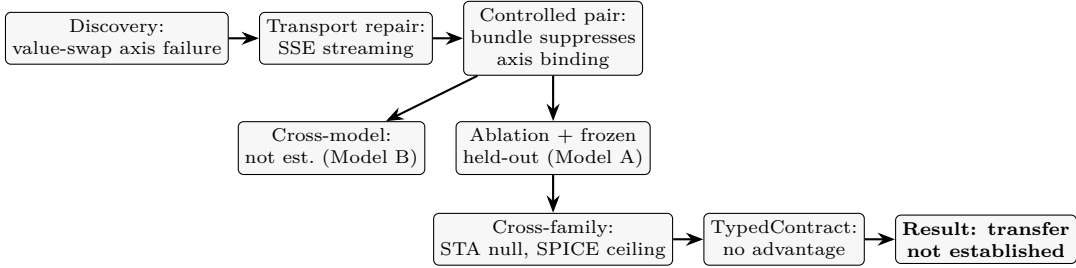


Figure 1: Study/evidence pipeline: three studies, the registration of each step, and where the central finding crystallizes.

2 Related Work and Problem Formulation

Harnesses and harness effects. A *harness* is the task-level information structure visible to the agent: the prompt, visible task files, disclosure bundle, public tool feedback, and the action surface (files the agent may edit). A *harness effect* is a change in agent behavior—specifically, semantic-binding correctness—attributable to a change in harness *information content*, holding the hidden truth, the grader, and the tool fixed.

Problem formulation. Each task presents a *semantic handoff*: the agent binds a tuple of values to canonical typed roles drawn from heterogeneous, role-misleading evidence. Formally, the agent recovers a binding $\beta = (r_1=v_1, \dots, r_m=v_m)$ over canonical roles $\{r_j\}$ from evidence whose surface role vocabulary is deliberately inconsistent with the canonical roles. Three families instantiate this: the *workflow-handoff* family binds PVT descriptors to typed axes; Family A (STA/PrimeTime) binds (intent_class, target_partition, check_mode) from an authority provenance DAG; Family B (SPICE/HSPICE) binds (corner, load_condition, metric) from a request–authority relational join. The external families share none of the workflow family’s signature structures (independent templates, role vocabularies, hidden-truth representations, graders, decoy logic); operational independence was verified under five pre-registered structural criteria (Appendix D), which we treat as operational independence, not a proof.

Positioning. Prior work establishes that harness configurations affect agent performance [2, 3, 4, 1, 5]. We instead study *when* a measured harness effect is scientifically attributable, *whether* it survives increasingly strict external-validity tests, and *which* execution-layer controls are required to support that conclusion. Table 1 sharpens this across five lines of related work.

Table 1: Positioning relative to five lines of related work (paraphrased).

Prior work	What it establishes	What this paper additionally does
Harness-Bench	Broad model×harness characterization	Controlled semantic-mechanism isolation + typed taxonomy; claim-validity (transport, membership, integrity)
Rethinking Harness Evaluation	Leaderboard search/overfitting	Even a <i>manually controlled</i> effect surviving pre-frozen within-family held-out may fail <i>cross-family</i> transfer
HarnessOpt-Bench	Optimizer <i>capability</i>	Studies the <i>validity</i> of harness-effect claims, not optimization
ABC / Best-Practices	Setup; shortcut/reward validity	Additionally audits transport, recovered degradation, sample membership, tool health, action surfaces, canonical-source integrity during execution
Protocol-validity work	Protocol design; reward-hacking surface	Treats measurement validity as the contribution; reports each control that <i>manufactured or masked</i> a result

3 Harness Effect Audit Methodology

Tasks and tool grounding. All three families run on real commercial tools (PrimeTime for the workflow and STA families; HSPICE for SPICE) through a transparent remote execution shim to a compute server hosting the tools—not simulations of tools. A wrong binding produces a plausible green signoff (STA) or plausible numeric measurement (SPICE), so tool success is necessary but not sufficient; only the typed provenance/authority grader rejects a wrong binding.

Conditions. Three conditions differ only in disclosure representation; hidden truth and grader are identical across conditions. **Base:** ambiguous natural-language handoff; no answer-bearing disclosure. **BundleS:** canonical role labels (C1), disjoint-axis value-domain definitions (C2), a glossary with references (C4), and a procedural contract (C7)—mapped mechanically from a frozen treatment specification; *no* answer-bearing assertion (no C6); *no* golden values. **TypedContract:** the same information content as BundleS as a machine-readable JSON Schema; no golden values; matched budgets.

Design. Seeded exact-counterbalanced blocked randomization, frozen before any paid call. Instance is the primary experimental unit; stochastic repetitions are nested observations; we report paired directions and raw counts per instance and do not pool trajectories into a single significance headline. Predeclared interpretation tables govern each contrast (Appendix B).

Detectability and the role of repetitions. With three primary instances per external family, our design is diagnostic rather than population-estimating. It is intended to test existence, recurrence, ceiling/null behavior, and transfer direction; it cannot resolve small effect sizes or support precise population-level success-rate estimates. We therefore do not present conventional powered population inference from three external task instances, and we do not use small- n bootstrap significance or a trajectory-pooled p -value to compensate for the limited instance count. The descriptive trajectory-stability analysis below reports what the data do support.

Trajectory stability (descriptive). Table 2 reports, for every main condition, the per-instance per-repetition semantic-binding outcome and within-instance agreement. The pattern is informative: across the twelve external condition-cells only one cell disagrees within its two repetitions (STA instance 0001 under Base); all SPICE cells and the remaining STA cells agree within instance. Run-to-run variance is thus low *within* instance—between-instance spread, not sampling noise, drives the family means. This is stability of direction at fixed instance, not a population estimate.

Table 2: Descriptive trajectory stability per main condition (✓=correct typed binding; ✗=semantic-binding failure). Reps are nested observations within instance. Agreement is within-instance across reps.

Condition	inst×rep	per-instance per-rep outcome	within-inst agree	raw
Workflow dev pair, Base (0009)	1×3	[✗✓✗]	disagree	1/3
Workflow dev pair, clear (0010)	1×3	[✓✓✓]	agree	3/3
Workflow held-out, Base (0011)	1×3	[✗✓✗]	disagree	1/3
Workflow held-out, BundleS (0017)	1×3	[✓✓✓]	agree	3/3
Workflow, Model B, Base	1×4	[✓✓✗✓]	disagree	3/4
Workflow, Model B, BundleS	1×4	[✗✓✓✓]	disagree	3/4
Family A (STA), Base	3×2	[✗✓] [✓✓] [✗✗]	dis./ag./ag.	3/6
Family A (STA), BundleS	3×2	[✗✗] [✓✓] [✗✗]	agree	2/6
Family B (SPICE), Base	3×2	[✓✓] [✓✓] [✓✓]	agree	6/6
Family B (SPICE), BundleS	3×2	[✓✓] [✓✓] [✓✓]	agree	6/6

Study registration. Sequential adaptation is a central concern for multi-stage agent research. Table 3 records the registration type of each result; only the predeclared, frozen-held-out, exact-counterbalanced, and cross-family-confirmatory results are headline.

Table 3: Study registration. Headline claims rest on predeclared / frozen-held-out / exact-counterbalanced / cross-family-confirmatory evidence; exploratory localization is not a headline.

Registration type	Result (used in)	Headline?
Predeclared (controlled pair)	Workflow family: clarity bundle suppresses axis binding (Study I)	Yes
Frozen-held-out confirmatory	Workflow family: BundleS on a pre-frozen held-out truth (Study I)	Yes
Exact-counterbalanced	Cross-model: BundleS vs Base under balanced blocking (Study II)	Yes
Cross-family confirmatory	STA null + SPICE ceiling on pre-registered families (Study II)	Yes
Secondary extension (pre-frozen)	TypedContract, separately frozen budget (Study II)	Yes (scoped)
Exploratory (sequential)	Component localization C1/C2/C4/C7/C24 (Appendix A)	No

4 Study I — Controlled Semantic-Binding Effects (RQ1)

Discovery (predeclared controlled pair). We construct a controlled pair with identical hidden truth and grader differing only in the disclosure bundle: an *ambiguous* handoff versus a *clear* handoff that adds the full clarity bundle. Both versions accept a green signoff regardless of binding correctness. The full clarity bundle suppresses the axis-binding failure for both models: Model A moves 1/3 (2 axis) → 3/3 (0 axis); Model B moves 0/3 (3 axis) → 3/3 (0 axis). The disclosure—not the tool, not the hidden truth—is the operative variable (RQ1: *established*).

The failure the bundle suppresses (Figure 2). Across the workflow-family ledger (54 primary episodes) the typed grader partitions outcomes into 29 correct typed bindings, 20 *axis-binding* failures

(a value on the wrong typed axis, e.g. `scenario=func` instead of `corner=func`), and 5 *role-conditioned value-selection* failures (type-valid, wrong value). The axis-binding subtype is exactly the failure the bundle targets: syntactically legal, tool-green, semantically wrong. Family-specific subtypes (coverage-cell mismatch / authority-unattested binding for STA; corner/load/metric authority misbind for SPICE) extend the taxonomy.

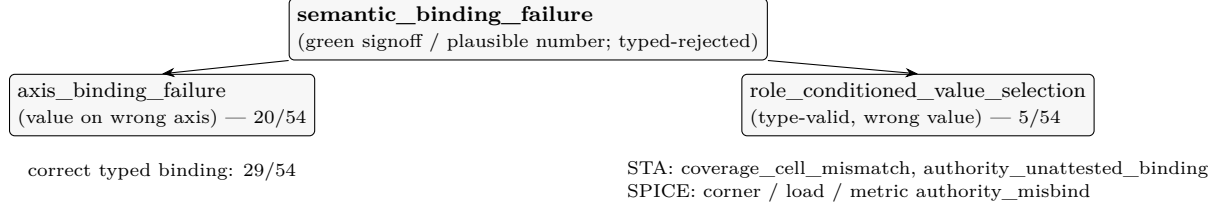


Figure 2: Semantic-binding failure taxonomy. All failures are tool-green and rejected only by the typed grader.

Non-answer localization. The full bundle includes an answer-bearing assertion (C6). Excluding C6, the schema/contract subset **BundleS** (C1+C2+C4+C7) suffices on the development pair for Model A (BundleS 3/3 vs the decoy-disambiguation bundle **BundleD** = C3+C5, which scores 1/3 and reproduces two wrong-axis bindings). The effect is attributable to the non-answer-bearing schema/contract disclosure—neither to leaking the answer (C6) nor to disambiguating decoys (C3/C5).

Pre-frozen held-out confirmation. Generalization within the family is tested on a held-out instance whose hidden truth was frozen *before* the disclosure mapping was selected: BundleS on the held-out truth yields 3/3 vs Base 1/3 on the matched Base instance. This is the strongest within-family evidence and the stepping-off point for external validity.

Model contingency. Under exact counterbalancing on Model B, BundleS is *not established*: Base 3/4 (1 axis) equals BundleS 3/4 (1 axis). The within-family benefit is model-contingent.

Component decomposition (exploratory; not headline). A sequential decomposition tested whether a stable minimal component underlies BundleS; no stable minimal component was identified (Appendix A, including a predeclared in-window C24 bridge that failed its replication threshold). The bundle-level effect is real within the family for Model A while a stable sub-component was not isolated.

The within-family story (Table 4, workflow rows): a non-answer-bearing disclosure suppresses a specific tool-green semantic failure, replicates on a pre-frozen held-out instance, but is already model-contingent under counterbalancing.

5 Study II — Model and Cross-Family External Validity (RQ2)

This study is where the central finding lives. We test transfer along two axes—*model* (a second model under exact counterbalancing) and *family* (two structurally independent families)—both evaluated with the model for which the within-family effect *was* established. The result is visible immediately in Table 4.

Cross-model (within family). Under exact counterbalancing, the within-family BundleS benefit is not established for Model B (tie $3/4 = 3/4$): the effect that replicated on the frozen held-out instance for Model A does not reproduce under balanced blocking.

Table 4: Main result matrix. Cells are semantic-binding correct rate; instance is the unit and repetitions are nested. “—” = not run.

Family	Model	Base	BundleS	TypedContract	Registration
Workflow (dev pair)	A	1/3 (2 axis)	3/3 (0 axis)	—	Predeclared
Workflow (frozen held-out)	A	1/3	3/3	—	Frozen held-out
Workflow (exact-counterbalanced)	B	3/4 (1 axis)	3/4 (1 axis)	—	Counterbalanced
Family A — STA / PrimeTime	A	0.50	0.33	0.33	Cross-family
Family B — SPICE / HSPICE	A	1.00	1.00	1.00	Cross-family

Cross-family — two distinct nulls, kept separate. *Family A (STA) — a clean null.* Across three instances BundleS does not improve over Base (0/3 improve; one declines 0.50→0.00, one ties at ceiling 1.00, one ties at floor 0.00). Family means: Base 0.50, BundleS 0.33, TypedContract 0.33. This is primary evidence of non-transfer on a family for which the measurement is discriminating. *Family B (SPICE) — non-discriminative at ceiling.* Base already solves the binding on all three instances (1.00), so BundleS and TypedContract also sit at 1.00. This is not a refutation; the effect is *unmeasurable* because Base leaves no room to improve. We do not merge the STA null and the SPICE ceiling into one “no transfer” claim.

Typed representation adds nothing. TypedContract renders the same non-answer-bearing information as BundleS in a machine-readable JSON Schema. Across the six external instances it produces no advantage over Base (0 improve, 1 decline, 5 tie) or BundleS (0/0/6 tie). We do not claim typed contracts are generally ineffective—only that, on the tested model and families, the representation does not restore transfer.

Central finding. *Within-family held-out confirmation did not imply cross-model or cross-family transfer.* The effect that replicated on a pre-frozen held-out instance was already model-contingent under counterbalancing and did not transfer to either external family in a way we could measure. Our instances test existence, recurrence, and transfer direction, not population-level pass rates; the STA null is consistent with “no effect” but, at three instances, cannot rule out a small effect; SPICE is a ceiling, not a refutation.

SPICE repair chronology (explicit). The SPICE result reflects a clean, audited chronology, not a post-hoc rewrite:

1. **Frozen Phase-5C primary collection** (Model A, 24 episodes, pre-frozen schedule/budget/conditions) produced the cross-family Base/BundleS data.
2. A SPICE action-surface integrity rule flagged a deck file as forbidden-edit on all 12 SPICE episodes, zeroing them.
3. A **no-call forensic audit** (no model calls) re-derived each deck from its submitted measurement configuration and diffed it against the shipped deck, establishing that the flagged change was the **intended regeneration of derived .param values**—0/12 protocol-compromised.
4. A subsequent **action-surface repair** (an immutable, integrity-hashed circuit core plus a derived deck) eliminated the false positive in a separately-frozen secondary collection.
5. A **separately-labeled Phase-5D secondary** three-condition replication (Base/BundleS/TypedContract, 36 episodes, Latin-balanced) ran under the repaired action surface (0 trips). Phase-5C results were **not rewritten or selectively replaced** after the repair; the Phase-5C primary remains frozen, and the repair is reported as an action-surface correction that made SPICE measurable (it was already at ceiling, so the semantic-binding rates are unchanged).

6 Study III — Evaluation and Measurement Validity (RQ3)

Each headline conclusion—positive (RQ1) and null (RQ2)—could have been manufactured or masked by an evaluation artifact. Study III abstracts our infrastructure into the **Harness Effect Audit Protocol**: an auditable evaluation protocol instantiated and stress-tested in this study, comprising four layers (Figure 3) operationalized as nine controls (Appendix C). We describe it as instantiated and stress-tested here; we do not claim broad external reusability or validation on unrelated benchmarks. For each layer we give the control and one concrete episode in which it caught a real threat that would otherwise have altered interpretation.

L1 Capability	tool-green alternatives; provenance/authority oracle; semantic failure taxonomy
L2 Sampling	repeated trajectories; blocking/counterbalancing; committed membership arbiter
L3 Execution	streaming transport; terminal-vs-recovered; tool-health bookends; protocol completion
L4 Artifact	action-surface isolation; exact-commit worktree + frozen hashes; custody + sanitized logs

Figure 3: The Harness Effect Audit Protocol (four layers, nine controls). Each layer is reported independently; none is collapsed into a single score.

L1 — Capability validity. Tool-green semantic alternatives, a provenance/authority oracle, and a typed failure taxonomy define *what is measured*. *Episode:* a Model A trajectory placed the descriptor on the wrong typed axis yet obtained a green PrimeTime signoff; the typed grader rejected it as `axis_binding_failure`. Without L1 this—and the 20 axis-binding failures in the ledger—would have been scored correct, erasing the very failure the bundle suppresses.

L2 — Sampling validity. Repeated trajectories, blocking/counterbalancing, and a committed sample-membership arbiter. *Episode:* the same controlled-pair instance is non-deterministic across stochastic trajectories (a wrong-axis outcome recurs in a minority of reps); the repeated-trajectory design with exact counterbalancing is what makes the directional claim defensible, and the arbiter (sole membership authority; `measurement_valid = terminal_transport_valid AND workspace_gradeable`) prevents a transport-invalid episode from being scored as a capability failure.

L3 — Execution validity. Streaming transport (terminal transport validity), terminal-validity-vs-recovered-degradation, tool-health bookends, protocol-completion tracking. *Episode:* during development Model A appeared to *fail* the controlled pair; the cause was transport, not capability—a non-streaming HTTP client censored the model’s long reasoning mid-generation, yielding an empty response misclassified as a capability failure. SSE streaming recovered the real behavior. This is why terminal transport validity is a measurement prerequisite, and why a gradeable episode that reached its answer only after recovered failed attempts is *kept*, not replaced. Tool-health bookends (a real-tool sentinel and a full golden-through-grader measurement control) gate each block so a tool-server outage is not misread as a model failure.

L4 — Artifact integrity. Action-surface isolation, exact-commit isolated worktrees with frozen canonical hashes, custody hashes with sanitized logs. *Episode:* a canonical golden configuration file was silently rewritten to an empty object by an unidentified external process, which a frozen-hash check misreported as a tool outage for roughly a day; the canonical-tree guard plus a git-status cross-check disambiguated the true cause. The SPICE action-surface false positive (Study II) was likewise an L4 threat, corrected by the immutable-core/derived-deck repair and a forensic audit.

What the protocol buys. Each layer is reported independently. The positive Study I conclusion required L1 (to see the failure), L3 (streaming, to not censor the model), and L2 (to not be misled by a single draw); the null Study II conclusion required L4 (so the SPICE ceiling was not a zeroing

false positive) and L2 (so transport-invalid episodes did not pass and erase the STA null). The stack is the set of controls without which neither claim is credible.

7 Limitations and Discussion

We are explicit about scope. *Controlled executable tasks, not sampled production incidents*: the tasks isolate the semantic-binding mechanism and are not a random sample of real chip-design incidents. *EDA-heavy environments*: all families are EDA tool-grounded; the failure mode is cognitive and tool-agnostic, but the external-validity result is established only within EDA. *Three external instances per family*: our design is diagnostic, not population-estimating; at three instances the STA null cannot rule out a small effect—we state “transfer not established,” not “refuted.” *Model-centered positive evidence*: the within-family effect is established for one model and not established for a second under counterbalancing; cross-family evaluation used the responsive model only. *Sequential exploratory localization*: the component decomposition was sequential and is not a headline. *Uncontrolled provider sampling seeds*: we mitigate with repeated trajectories and counterbalancing and report raw counts. *SPICE ceiling*: one external family is non-discriminative at ceiling. *No stable minimal bundle component*. *TypedContract* scoped to the tested model and families. Separately, the external families were deliberately constructed to be structurally independent, so cross-family non-transfer is partly the expected behavior of the construction; the cleaner transfer evidence is the same-family, cross-model tie.

Discussion. The result is a caution and a method. The caution: a single within-family replication—even a frozen held-out one—is weak evidence of generalization for harness effects, which can be model- and family-specific. The method: the audit protocol makes the positive and null conclusions independently checkable. We do not claim harness effects never generalize—only that *this* effect did not generalize across the tested model and families, and that generalization deserves the same audit as the effect itself.

Reproducibility. Every result is reproducible from frozen manifests: deterministic generators with fixed seeds; independent graders per family; an exact-commit isolated-worktree execution with pre/post-episode canonical-hash verification (canonical-tree integrity guard); committed sample-membership arbitration; validity-only replacement; per-episode custody byte-matching; and sanitized custody evidence. Frozen manifests, the pre-registered treatment mapping, analysis schedules, and phase provenance are in the supplementary material (anonymized). Program totals: 58 (workflow) + 24 (cross-family core) + 36 (cross-family secondary) = 118 primary paid episodes; committed-ledger cost ¥701.72.

8 Conclusion

A harness intervention can survive repeated development experiments and a pre-frozen within-family held-out test, yet disappear under a new model or task family. We showed this concretely: a non-answer-bearing clarity bundle suppresses a specific tool-green semantic-binding failure within a family (confirmed on a pre-frozen held-out instance), is model-contingent under exact counterbalancing, and does not transfer to two independent families in a way we could measure; a typed representation adds no advantage. Because each conclusion could be manufactured or masked by an artifact, we contributed the Harness Effect Audit Protocol (four layers, nine controls). The contribution is an evaluation study—of *when* an effect is attributable, *whether* it generalizes, and *which* controls make either claim credible—not a proposal of an improved harness.

References

- [1] ABC authors. Establishing best practices in building rigorous agentic benchmarks (ABC), 2025. VERIFICATION-REQUIRED: placeholder metadata; replace with verified canonical citation.

- [2] Harness-Bench authors. Harness-Bench: Broad characterization of model×harness interactions in LLM agents, 2025. VERIFICATION-REQUIRED: placeholder metadata; replace with verified canonical citation.
- [3] Harness-Evolution authors. Rethinking the evaluation of harness evolution in LLM agent benchmarks, 2025. VERIFICATION-REQUIRED: placeholder metadata; replace with verified canonical citation.
- [4] HarnessOpt-Bench authors. HarnessOpt-Bench: Optimizing agent harness configurations, 2025. VERIFICATION-REQUIRED: placeholder metadata; replace with verified canonical citation.
- [5] Protocol-Validity authors. Protocol validity and reward-hacking in LLM agent benchmarks, 2025. VERIFICATION-REQUIRED: placeholder metadata; replace with verified canonical citation.

A Component decomposition (exploratory)

The C1/C2/C4/C7/C24 sequential localization, including a predeclared in-window C24 bridge that failed its $\geq 3/4$ -with-0-axis replication threshold (**not_established**); corrected Stage-3 causal-scope wording (directional evidence for a candidate joint effect; not a formally identified interaction). Headline: no stable minimal component isolated.

B Per-episode tables and predeclared interpretation rules

Instance-level paired directions and raw counts for the workflow family (54 ledger episodes: 29 correct / 20 axis / 5 role-conditioned-value), the cross-model exact-counterbalanced block, and the three-by-two STA/SPICE external instances (Phase-5D three-condition rows included); the frozen interpretation thresholds for each contrast. A clearly-labeled instance-level bootstrap sensitivity (over the three external instances) appears here only as a sensitivity check and does not drive any headline claim or p -value.

C Nine-control operationalization and infrastructure

The four-layer→nine-control mapping; the canonical-tree integrity guard (**freeze/verify/enforce**, **FAILED_INTEGRITY** stop, regression tests); the episode arbiter and validity-only replacement rule; the streaming transport control; tool-health sentinel and full-path measurement bookends; the immutable-core/derived-deck action-surface repair with regressions; custody hashing and log-sanitization rules.

D Operational independence and phase provenance

The five operational-independence criteria for the external families; the internally-gated development-stage history as a provenance record, including episode counts (workflow 58; cross-family core 24; cross-family secondary 36) and committed-ledger costs (workflow ¥682.25; cross-family core ¥7.72; cross-family secondary ¥11.75; total ¥701.72). Exact commit hashes and custody manifests are provided in anonymized supplementary material.

E Threats caught by each layer (episode log)

The concrete episodes referenced in Study III, each with the control that caught it and the interpretation that would otherwise have changed.

AI-use statement (outside the main-text page limit). During preparation, an LLM-based coding and writing assistant was used for experiment orchestration code, deterministic task generators, evaluation infrastructure, report scripts, and prose drafting. All experimental results are produced by committed, deterministic pipelines and reproduced from frozen manifests; no scientific claim was authored or validated by an LLM acting as an oracle. All numerical results were generated by committed code and re-extracted from committed ledgers.

Reproducibility statement. Deterministic generators (fixed seeds), independent per-family graders, exact-commit isolated-worktree execution with pre/post-episode canonical-hash verification, committed sample-membership arbitration, validity-only replacement, per-episode custody byte-matching, and sanitized custody evidence are provided. Frozen manifests, the treatment mapping, schedules, and provenance are in anonymized supplementary material.

Anonymous code/artifact instructions. Code, frozen manifests, deterministic generators, evaluators, and sanitized custody evidence are provided as anonymized supplementary material. Models are public and are referred to as Model A / Model B; compute-host and shim details are generalized. The artifact does not require any non-public resource to reproduce the reported numbers.